

Graduation Project

Final Report File

Implementing VPN Solutions with FortiGate

**Fortinet Cybersecurity Engineer
(ONL3_ISS8_S5)**

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1.Introduction

1.1 Document Purpose:

The purpose of this document is to:

- Provide a comprehensive overview of implementing secure VPN solutions on FortiGate devices.
- Explain the concepts behind different VPN technologies.
- Document the configuration steps for SSL VPN and IPsec VPN tunnels.
- Demonstrate how SD-WAN features can enhance VPN traffic performance.
- Show practical steps, configurations, performance tests, and results.

1.2 Key Features:

1. Implementation of Multiple VPN Technologies

Covers SSL VPN for remote user access and IPsec VPN for secure site-to-site connectivity.

2. FortiGate Configuration

s Provides practical experience with FortiGate firewalls, including policy creation, authentication setup, and tunnel configuration.

3. SD-WAN Integration for Traffic Optimization

Demonstrates how FortiGate's SD-WAN features can improve VPN performance through intelligent path selection.

4. Performance Testing

Includes connectivity tests, throughput checks, and verification of VPN stability and functionality.

5. Reporting

Producing configuration guides, test results, SD-WAN performance reports.

2. Network Architecture

2.1 Topology description:

This topology represents a **dual-site network setup** where two separate LAN environments (left and right) connect to a shared external network (“Net”) through individual **Fortinet firewalls** and **Cisco 3725 edge routers**.

At the top center of the topology:

- A cloud labeled “**Net**” represents the **public network or WAN (possibly the Internet or a private WAN link)**.
- The cloud connects to **two separate Fortinet firewall appliances** (port2 on both devices), meaning each site has its own dedicated firewall to reach the WAN.

Perimeter Layer

- The Fortinet firewalls is connected on **port1** to a **Cisco 3725 router** via its **fa0/1** interface.
- The router then uses **fa0/0** to connect downstream into the LAN through another router.

3. LAN Distribution Layer

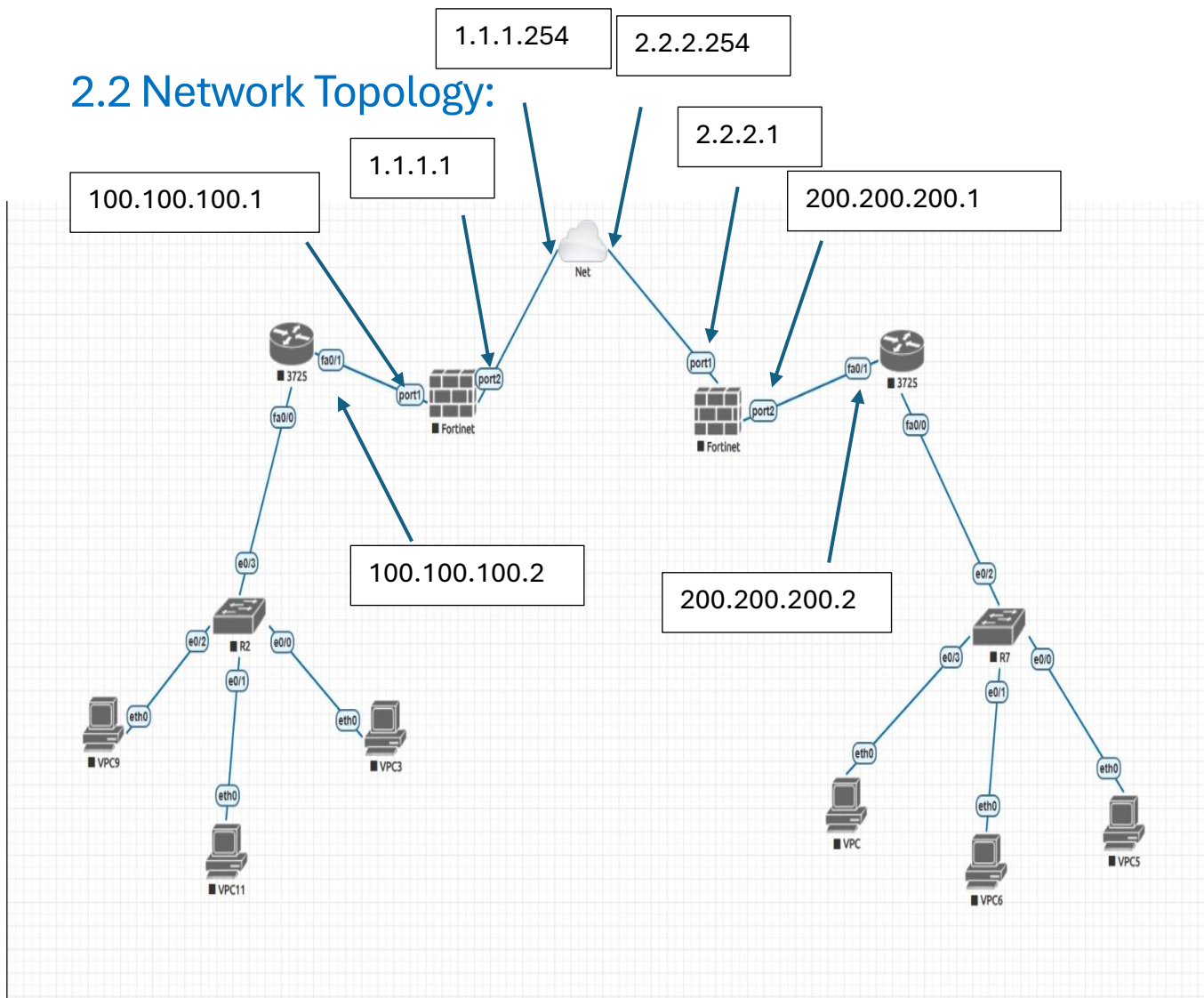
- Router **R2 and R7** distributes the connection into three segments using interfaces **e0/0, e0/1, and e0/2**.

4. End Devices

Connected to R2:

- **VPC3** and **VPC5** on interface e0/0
- **VPC11** and **VPC 6** on interface e0/1
- **VPC9** and **VPC** on interface e0/2

2.2 Network Topology:



3.Naming Convention:

We will adopt a standardized naming convention for network devices to simplify identification, management, and troubleshooting, using a consistent naming schema across all infrastructure components.

Hostname	IP address
R1	192.168.1.1
SW1	192.168.1.2
VPC3	Dynamic address
VPC9	Dynamic address
VPC11	Dynamic address
R2	192.168.2.1
SW2	192.168.2.2
VPC4	Dynamic address
VPC5	Dynamic address
VPC6	Dynamic address

-for any switch and router used the username is ahmed and the password is ccna.

-Network Subnet mask: 255.255.255.0

4. Network Design Notes and Configuration

4.1 VPN concepts and configuration:

4.1.1 Types of VPNs and use cases:

1-SSL VPN (Secure Sockets Layer VPN)

Description:

- Works through a web browser or VPN client.
- Uses HTTPS (TCP 443), making it easy to access from anywhere.
- Does not require full-tunnel installation for web-based portal access.

Use Cases:

- Remote employees accessing company applications securely.
- Users connecting from public networks (cafes, airports).
- BYOD environments where users connect from personal devices.
- Accessing internal portals, file servers, or RDP over a secure tunnel.

2-IPsec VPN (Internet Protocol Security VPN)

Description:

- Network-layer VPN that encrypts traffic between two endpoints.
- Can be used for **site-to-site** or **remote access**.
- Uses strong encryption standards (IKEv1, IKEv2).

Use Cases:

- Secure connection between branch offices and headquarters.
- Permanent tunnel between two networks at different locations.
- Connecting data centers or cloud networks.
- High-security environments.

4.1.2 SSL VPN configuration on FortiGate:

Overview:

SSL VPN on FortiGate allows remote users to securely access internal network resources through an encrypted SSL/TLS tunnel. FortiGate supports two SSL VPN modes: **Web Portal Mode** and **Tunnel Mode**, making it flexible for both browser-based access and full network connectivity.

Configuration:

4.1.3 Testing and Documenting VPN functionality:

Overview:

Testing VPN functionality ensures that the VPN is correctly configured, secure, and performing as expected. Below is a structured overview covering the essential areas of VPN testing.

4.1.4 VPN configuration guide:

4.2 IPsec VPN Configuration

4.2.1 Understanding IPsec architecture and phases:

IPsec has **three main architectural components**:

1-Security Protocols

AH (Authentication Header)

ESP (Encapsulating Security Payload)

2- Security Associations (SAs)

An SA is a set of rules and cryptographic keys that define how traffic is protected.

3- Key Management (IKE)

IKE (Internet Key Exchange) is used to create and manage IPsec SAs.

4.2.2 Creating an IPsec tunnel between two FortiGate devices:

Configuration:

4.2.3 Connectivity testing and troubleshooting:

Overview:

An IPsec VPN ensures secure communication between two networks over the internet. To verify its proper operation, connectivity testing and troubleshooting help identify issues in tunnel negotiation, routing, encryption, and access control.

4.2.4 IPsec configuration document and test results:

Configuration Guide:

4.3 FortiGate VPN with SD-WAN

4.3.1 Introduction to SD-WAN concepts:

Software-Defined Wide Area Networking (SD-WAN) is a modern networking technology that simplifies the management, performance, and reliability of WAN connections. In traditional WAN setups, traffic is usually routed through fixed physical circuits such as MPLS or a single ISP link. SD-WAN enhances this by using **software-based intelligence** to dynamically control how traffic flows across multiple WAN links.

4.3.2 Configuring SD-WAN on FortiGate:

Configuration:

4.3.3 Optimizing VPN traffic using SD-WAN policies:

Overview:

SD-WAN enhances traditional IPsec VPN setups by intelligently routing traffic across multiple WAN links based on real-time performance. Instead of forcing all VPN traffic through a single ISP link, SD-WAN continuously measures link quality and automatically chooses the best path for the VPN tunnel or specific applications.

This ensures higher reliability, better performance, and reduced downtime for site-to-site communication.

4.3.4 Connectivity Testing:

Overview:

Connectivity testing ensures that these features work together effectively, providing reliable and high-performance communication across the network.

Device Pinging:

